



CCIE Enterprise Infrastructure (v1.0) Training Topics

Network Infrastructure

1.1 Switched campus

EtherChannel

- i LACP, Static
- ii Layer 2, Layer 3
- iii Load balancing
- iv EtherChannel Misconfiguration Guard

Spanning Tree Protocol

- i PVST+, Rapid PVST+, MST
- ii Switch priority, port priority, path cost, STP timers
- iii PortFast, BPDU Guard, BPDU Filter
- iv Loop Guard, Root Guard

1.2 Routing Concepts

- i Route filtering with any routing protocol
- ii Manual summarization with any routing protocol
- iii Redistribution between any pair of routing protocols
- iv Routing protocol authentication
- v Bidirectional Forwarding Detection

1.3 EIGRP

- a Adjacencies
- b Best path selection
 - i. RD, FD, FC, successor, feasible successor
 - ii. Classic Metrics and Wide Metrics
- c EIGRP Named Mode
- d Optimization, convergence and scalability



Availability of
CCIE Bootcamps
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- i Fast convergence requirements
- ii Leak-map with summary routes
- iii EIGRP stub with leak map

1.4 OSPF (v2 and v3)

1.4.a Adjacencies

1.4.b Network types, area types

1.4.c Path preference

1.4.d Operations

1.4.d i General operations

1.4.e Optimization, convergence and scalability

1.4.e i Metrics

1.4.e ii LSA throttling, SPF tuning, fast hello

1.4.e iii LSA propagation control (area types)

1.4.e v Prefix suppression

1.5 BGP

1.5.a IBGP and EBGP peer relationships

1.5.a ii Private AS

1.5.b Path selection

1.5.b i Attributes

1.5.b ii Best path selection algorithms

1.5.b iii Load balancing

1.5.c Routing policies

1.5.c i Attribute manipulation

1.5.c ii Outbound Route Filtering

1.5.c iii Standard and extended communities

1.5.c iv Multi-homing

1.5.d. i local-AS, allowas-in, remove-private-as

1.5.d ii Prepend

1.5.d iii Regexp



- 1.5.e Convergence and scalability
 - 1.5.e i Route reflector
 - 1.5.e ii Aggregation, as-set

- 1.5.f Other BGP features
 - 1.5.f i Multipath, add-path

1.6 Multicast

- 1.6.a Layer 2 multicast
 - 1.6.a i IGMPv2, IGMPv3
 - 1.6.a ii IGMP Snooping, PIM Snooping
 - 1.6.a iii IGMP Querier
- 1.6.b Reverse path forwarding check
- 1.6.c PIM
 - 1.6.c i Sparse Mode
 - 1.6.c ii Static RP, BSR,
 - 1.6.c iii Group to RP Mapping
 - 1.6.c iv Bidirectional PIM, PhantomRP

2. Software Defined Infrastructure

2.1 Cisco SD Access

- 2.1.a Design a Cisco SD Access solution
 - 2.1.a i Underlay network (IS-IS, manual/PnP)
 - 2.1.a ii Overlay fabric design (LISP, VXLAN, Cisco TrustSec)
 - 2.1.a iii Fabric domains (single-site and multi-site using SD-WAN transit)
- 2.1.b Cisco SD Access deployment
 - 2.1.b i Cisco DNA Center device discovery and device management
 - 2.1.b ii Add fabric node devices to an existing fabric
 - 2.1.b iii Host onboarding (wired endpoints only)
 - 2.1.b iv Fabric border handoff
- 2.1.c Segmentation
 - 2.1.c i Macro-level segmentation using VNs



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2.2 Cisco SD-WAN

2.2.a Design a Cisco SD-WAN solution

2.2.a i Orchestration plane (vBond, NAT)

2.2.a ii Management plane (vManage)

2.2.a iii Control plane (vSmart, OMP)

2.2.a iv Data plane (vEdge/cEdge)

2.2.b WAN edge deployment

2.2.b i Onboarding new edge routers

2.2.b ii OMP

2.2.b iii TLOC

2.2.c Configuration templates

2.2.d Localized policies

2.2.e Centralized policies

3. Transport Technologies and Solutions

3.1 MPLS

3.1.a Operations

3.1.a i Label stack, LSR, LSP

3.1.a ii LDP

3.1.a iii MPLS ping, MPLS traceroute

3.1.b L PN 3V

3.1.b i PE-CE routing

3.1.b ii MP-BGP VPNv4/VPNv6

3.1.b iii Extranet (route leaking)

VRF-lite

3.2 DMVPN

3.2.a Troubleshoot DMVPN Phase 3 with dual-hub

3.2.a i NHRP

3.2.a ii IPsec/IKEv2 using pre-shared key



3.2.a iii Per-Tunnel QoS

3.2.b i Site-to-site, Server, Client, Spoke-to-Spoke

3.2.b ii IPsec/IKEv2 using pre-shared key

4. Infrastructure Security and Services

4.1 Device Security on Cisco IOS XE

4.1.a Control plane policing and protection

4.2 Network Security

4.2.a iii DHCP Snooping, DHCP option 82

4.2.a vi Port Security

4.2.a vii Private VLAN

4.2.b Router security features

4.2.b i IPv6 Traffic Filters

4.2.b ii IPv4 Access Control Lists

4.2.c IPv6 infrastructure security features

4.2.c i RA Guard

4.4 Quality of Service

4.4.a End to End L3 QoS using MQC

4.4.a i DiffServ

4.4.a ii CoS and DSCP Mapping

4.4.a iii Classification

4.4.a iv Marking using IP Precedence, DSCP, CoS

4.4.a v Policing, shaping

4.4.a vi Congestion management and avoidance



4.5 Network Services

4.5.a First Hop Redundancy Protocols

4.5.a i HSRP, GLBP, VRRP

4.5.a ii Redundancy using IPv6 RS/RA

4.5.b DHCP on Cisco IOS

4.5.b i Client, server, relay

4.5.b ii Options

4.5.b iii SLAAC/DHCPv6 interaction

4.5.b iv Stateful DHCPv6

4.5.d IPv4 Network Address Translation

4.5.d i Static NAT

4.5.d ii Dynamic NAT, PAT

4.6 Network optimization

4.6.a IP SLA

4.6.a i ICMP probes

4.6.b Tracking object

5. Infrastructure Automation and Programmability

5.1 Data encoding formats

5.1.a JSON

5.1.b XML

5.2 Automation and scripting

5.2.a EEM applets

5.2.b Guest shell

5.2.b i Linux environment

5.2.b ii CLI Python module

5.2.b iii EEM Python module



5.3 Programmability

5.3.a Interaction with vManage API

5.3.a i Python requests library and Postman

5.3.a ii Monitoring endpoints

5.3.a iii Configuration endpoints

5.3.b Interaction with Cisco DNA Center API

5.3.b i HTTP request (GET, PUT, POST) via Python requests library and Postman

5.3.c Interaction with Cisco IOS XE API

5.3.c i Via NETCONF/YANG using Python ncclient library

5.3.c ii Via RESTCONF/YANG using Python requests library and Postman

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